Explicit Littlewood-type bounds for $L(1,\chi)$ and class numbers from a zero-free half-plane

Jizhou Guo
mitsuha2021b@gmail.com

28 October 2026

Abstract

Let $1/2 \le \theta < 1$ and let χ be a nonprincipal Dirichlet character modulo $q \ge 3$ whose L-function has no zeros in $\operatorname{Re} s > \theta$. We prove, with every constant explicit, that

$$\frac{\zeta(2)e^{-\gamma}a\,e^{-\Theta}}{M+\log\log q} \le |L(1,\chi)| \le \frac{e^{\gamma}e^{\Theta}(M+\log\log q)}{a}, \qquad a = \tfrac{22}{25}(1-\theta),$$

for every $M \ge 5$, where $\Theta = \Theta(\theta, M, q)$ is an explicit small quantity ($\Theta < 0.0554$ for $\theta = 7/8$, $M = 10$). For $\theta = 7/8$, the half-plane asserted for all Dirichlet L-functions by OpenAI's paper of 30 September 2026, this gives

$$\frac{1}{10.41\,(10+\log\log q)} \le |L(1,\chi)| \le 17.12\,(10+\log\log q),$$

and, for every fundamental discriminant $-d < 0$, $h(-d) \ge \sqrt{d}/(32.69\,(10+\log\log d))$; in particular $h(-d) > 100$ for $d \ge 1.83\cdot 10^{9}$ and $h(-d) > 1000$ for $d \ge 1.88\cdot 10^{11}$. The shape of the bounds is Littlewood's (1928, under GRH, for quadratic characters); that it persists under a zero-free half-plane, with the width entering the constants, is stated by Friedlander (1977) as an easy extension of Littlewood's method. The explicit constants given here are unconditional exactly when the input theorem is correct. A Hecke–Landau-type bound $L(1,\chi) \ge 1/(120\log q)$ for real χ is also proved from the exclusion of real zeros in $(1-1/(40\log q), 1)$ alone. The constants are checked by the scripts in the ancillary files.

1 Introduction

Let χ be a nonprincipal Dirichlet character modulo $q \ge 3$ and write $s = \sigma + it$. Throughout, $\log L(s,\chi)$ denotes the branch obtained by analytic continuation of the absolutely convergent series $\sum_{n\ge 2}\Lambda(n)\chi(n)n^{-s}/\log n$ from $\operatorname{Re} s > 1$; on any simply connected region containing $\operatorname{Re} s > 1$ on which $L(s,\chi)$ has no zeros this branch is single-valued. We consider the hypothesis

$$(\mathrm{H}_\theta) \qquad L(s,\chi) \ne 0 \quad \text{for } \operatorname{Re} s > \theta, \tag{1.1}$$

for a fixed $\theta \in [1/2, 1)$. For $\theta = 1/2$ it is the Generalised Riemann Hypothesis for χ. For $\theta = 7/8$ and every Dirichlet character it is Theorem 1.1 of OpenAI's paper [1], which states that every Dirichlet L-function, including $\zeta(s)$, has no zero in $\operatorname{Re} s > 7/8$, poles of principal characters being allowed; the Lean scope document [2] describes a formalization of that statement for all moduli and all characters. Smaller boundaries have since been stated: $\sigma_\dagger = 0.8749570194\ldots$ in [3], and the Quasi-Riemann Tracker [4] records $4369779/5000000 = 0.8739558$ for Dirichlet L-functions. None

of these inputs is verified in this paper; every statement below is of the form "(H_θ) implies…" with explicit constants, so that it is unconditional exactly when the input theorem is correct.

Under GRH, Littlewood [5] proved that $L(1,\chi_d)$, for a quadratic character χ_d, is of size $(\log\log|d|)^{\pm1}$, with the constants $(\frac12+o(1))\zeta(2)e^{-\gamma}$ and $(2+o(1))e^{\gamma}$ (as quoted in [6, (1.1)]). Friedlander [7, (1)], writing for the Legendre symbol modulo a prime q, states as a theorem of Littlewood that if $L(s,\chi)\neq0$ for $\operatorname{Re}s>1-\theta(q)$ then $c_1\theta(q)/\log\log q<L(1,\chi)<c_2\log\log q/\theta(q)$ with positive absolute constants c_1,c_2, and remarks that Littlewood proved this only for $\theta(q)=1/2$ but that his method extends easily (as remarked by Elliott for the lower bound); the width enters to the first power there, as it does here through a. The modern form of the underlying approximation is the lemma of Granville and Soundararajan [6, Lemma 2.1], which under a zero-free rectangle expresses $\log L(s,\chi)$ by a short sum over prime powers. Explicit constants have been given under GRH by Lamzouri, Li and Soundararajan [8] for primitive χ with $q\geq10^{10}$, extended to all $q\geq3$ (class numbers: $q\geq5$) by Languasco and Trudgian [9]. OpenAI's paper of 5 October 2026 [10, §1] notes that Littlewood's short Euler-product argument applied to a fixed zero-free half-plane gives $h(D)\gg\sqrt{|D|}/\log\log|D|$ with a computable implied constant, without giving one. Kimura [11, v2, Remark 11], in a preprint (not yet refereed) that determines all imaginary quadratic fields with class number at most 2000 by making the Landau–Siegel exclusion of [12] explicit, notes that under such a half-plane Littlewood's argument gives $h(-d)\gg\sqrt{d}/\log\log d$ with an effective implied constant, as observed in [10, §1], and writes that he has not made it explicit under the half-plane. Here we work out explicit constants.

Constants

The constants below are functions of θ and of a free parameter $M\geq5$; their definitions are collected in Definition 2.6 and their values for several θ in Table 1. We write γ for Euler's constant and $\zeta(2)=\pi^2/6$.

Theorem 1.1. *Let $1/2\leq\theta<1$, $\delta=1-\theta$ and $a=22\delta/25$. Let χ be a nonprincipal Dirichlet character modulo $q\geq3$ satisfying (H_θ). Let $\kappa,T_0,A_1,A_2,A_3,K_0^+,J^+$ be as in Definition 2.6.*

(i) *For $1-a\leq\sigma\leq3+a$ and all real t,*

$$|\log L(\sigma+it,\chi)|\leq A_1\log q+A_2\log(|t|+T_0)+A_3.$$

(ii) *Let $M\geq5$, $y=(e^M\log q)^{1/a}$ and $\mathcal{L}=\log y=(M+\log\log q)/a$. Then*

$$\log L(1,\chi)=\sum_{p\leq y}-\log\Bigl(1-\frac{\chi(p)}{p}\Bigr)+\varepsilon,\qquad |\varepsilon|\leq E_0(q)+\frac{1.25506}{\mathcal{L}}+\frac{2.5}{\sqrt{y}}+\frac{\mathcal{L}}{y^{2/3}},$$

where $E_0(q)=\bigl(A_1K_0^+ + (A_2J^+ + A_3K_0^+)/\log q\bigr)/(2\pi e^M)$ and each $-\log(1-\chi(p)/p)=\sum_{k\geq1}\chi(p)^k/(kp^k)$.

(iii) *With $\Theta(q)=E_0(q)+1.25506/\mathcal{L}+2.5/\sqrt{y}+\mathcal{L}/y^{2/3}+1/(2\mathcal{L}^2-1)+2/y$,*

$$\frac{\zeta(2)e^{-\gamma}\,a\,e^{-\Theta(q)}}{M+\log\log q}\leq|L(1,\chi)|\leq\frac{e^{\gamma}\,e^{\Theta(q)}\,(M+\log\log q)}{a}.$$

$\Theta(q)$ is decreasing in q, so both bounds hold with $\Theta(3)$ in place of $\Theta(q)$. For real χ the bounds hold for $L(1,\chi)$ itself, which is positive.

The restriction $\theta \geq 1/2$ costs nothing: for $\theta < 1/2$ the hypothesis is never satisfied, since $L(s,\chi)$ has infinitely many zeros in the critical strip and the functional equation pairs a zero ρ with $1-\bar{\rho}$. Primitivity of χ is not needed; for an imprimitive χ the hypothesis is inherited from the primitive character inducing it, because the missing Euler factors have their zeros on $\operatorname{Re} s = 0$.

For $\theta = 7/8$ and $M = 10$ one has $a = 0.11$, $\Theta(3) = 0.05534\ldots$, $e^{\gamma}e^{\Theta(3)}/(a\zeta(2)) = 10.4034\ldots$ and $e^{\gamma}e^{\Theta(3)}/a = 17.1129\ldots$, whence:

Corollary 1.2. *Let χ be a nonprincipal character modulo $q \geq 3$ with $L(s,\chi) \neq 0$ for $\operatorname{Re} s > 7/8$. Then*

$$\frac{1}{10.41\,(10+\log\log q)} \leq |L(1,\chi)| \leq 17.12\,(10+\log\log q),$$

and, for $0.89 \leq \sigma \leq 3.11$ and all t, $|\log L(\sigma+it,\chi)| \leq 26.64\log q + 222\log(|t|+11.58) + 139.37$.

Corollary 1.3. *Assume that $L(s,\chi_{-d}) \neq 0$ for $\operatorname{Re} s > 7/8$ for every fundamental discriminant $-d < 0$, where χ_{-d} is the Kronecker symbol. Then for every fundamental discriminant $-d < 0$ the class number of $\mathbb{Q}(\sqrt{-d})$ satisfies*

$$h(-d) \geq \frac{\sqrt{d}}{32.69\,(10+\log\log d)}.$$

In particular $h(-d) > 1$ for $d \geq 1.67\cdot 10^5$, $h(-d) > 10$ for $d \geq 1.76\cdot 10^7$, $h(-d) > 100$ for $d \geq 1.83\cdot 10^9$ and $h(-d) > 1000$ for $d \geq 1.88\cdot 10^{11}$. Choosing M depending on the target improves these thresholds to $1.10\cdot 10^5$, $1.13\cdot 10^7$, $1.15\cdot 10^9$ and $1.17\cdot 10^{11}$ (Table 2).

The hypothesis of Corollary 1.3 is contained in Theorem 1.1 of [1]. The same proof gives $h(-d) \geq \sqrt{d}/(32.41\,(10+\log\log d))$ from the tracker boundary 0.8739558 (Table 1); the gain from the smaller boundaries is below one per cent because the constant is essentially $e^{\gamma}/(\zeta(2)a)$ with a proportional to the width $1-\theta$.

We also record the explicit form of the older Hecke–Landau argument, which needs only the absence of real zeros close to 1 and is proved by a different method (an auxiliary function with nonnegative coefficients).

Theorem 1.4. *Let $q \geq 3$, let χ be a real nonprincipal character modulo q, and put $\eta = 1/(40\log q)$. If $L(\sigma,\chi) \neq 0$ for every real σ with $1-\eta < \sigma < 1$, then $L(1,\chi) \geq 1/(120\log q)$. The hypothesis follows from (H_θ) whenever $\theta \leq 1 - 1/(40\log 3) = 0.9772\ldots$; in that case $h(-d) \geq \sqrt{d}/(120\pi\log d)$ for every fundamental discriminant $-d < 0$.*

For $q \geq 3$ the bound of Corollary 1.2 is stronger than $1/(120\log q)$ (the two agree at $\log q = 0.853$); Theorem 1.4 is included because its hypothesis is weaker and because it is the explicit form of the implication "no real zero in $(1 - c/\log q, 1) \Rightarrow L(1,\chi) \gg 1/\log q$" of Hecke (Landau [13]), with $c = 1/40$. Kimura [11, Lemma 8, Proposition 9] gives another explicit form of that implication, which he applies with gaps $c \approx 0.0055$–0.006 (his Table 1) and, under a half-plane, with $c \approx 0.6$ (his Remark 11); the Lean statement described in [2] gives such a gap with an unspecified c, so Theorem 1.4 does not follow from it, while it does follow from the 7/8 half-plane.

Method

The analytic route is the classical one: a Borel–Carathéodory bound for $\log L$ on discs inside the zero-free half-plane (Section 3), a smoothed Perron integral moved to $\operatorname{Re} s = 1-a$, which expresses $\log L(1,\chi)$ by the Euler product over $p \leq y = (e^M\log q)^{1/a}$ with an error that is explicitly small

(Section 4), and the explicit Mertens product bounds of Rosser and Schoenfeld (Section 5). The only choices are the geometry of the discs, which fixes $a = 22\delta/25$ and the Borel–Carathéodory factor $\kappa = 25/\delta + 22$, and the parameter M, which trades the size of the Perron error against the additive constant in $M + \log\log q$. The bound for $|L(s,\chi)|$ on the discs is obtained by partial summation with the trivial estimate for character sums, not by convexity, which costs a factor of about two in the exponent of q but keeps the proof self-contained; the loss is absorbed by e^M. The constant $e^\gamma/(\zeta(2)a)$ in the lower bound compares with Littlewood's $2e^\gamma/\zeta(2)$ for $\theta = 1/2$ (where $1/a = 25/11$), so the method loses only the factor $25/22$ and the additive M against the GRH result. A first version of the main theorem with weaker constants (19.1 and 61 in place of 10.41 and 32.69) was checked by an independent AI review (see the statement on the use of AI at the end); the improvements here come from treating the prime powers through the Euler product and from using Theorem 7 instead of Theorem 5 of [14].

Section 6 compares the thresholds with the unconditional determinations of Watkins [15] ($h \le 100$) and Kimura [11] ($h \le 2000$). It also explains why no computable data can contradict Corollary 1.3: for $d < 1.83 \cdot 10^9$ the bound is below 101, so there the corollary follows from Watkins's theorem and a direct computation. Section 7 proves Theorem 1.4, and Section 8 describes the ancillary scripts.

2 Inputs and constants

2.1 Classical inputs

The following facts are used in the form stated.

Lemma 2.1 (partial summation)**.** *Let χ be a nonprincipal character modulo q and $S(x) = \sum_{n\le x}\chi(n)$. Then $|S(x)| \le q$ for all x, $S(q) = 0$, and for $\sigma > 0$*

$$L(s,\chi) = \sum_{n\le q}\chi(n)n^{-s} + s\int_q^\infty S(x)x^{-s-1}\,\mathrm{d}x. \tag{2.1}$$

Consequently, for $0 < \sigma_0 < 1$ and $\sigma \ge \sigma_0$,

$$|L(s,\chi)| \le \frac{q^{1-\sigma_0}}{1-\sigma_0} + \frac{|s|\,q^{1-\sigma_0}}{\sigma_0}. \tag{2.2}$$

Proof. S vanishes at multiples of q and has at most q nonzero terms in any incomplete period, so $|S(x)| \le q$. For $\sigma > 1$, partial summation of $\sum_{n>q}\chi(n)n^{-s}$ against S gives (2.1), the boundary term being $-S(q)q^{-s} = 0$; the integral converges absolutely and is analytic for $\sigma > 0$, so (2.1) holds there. For (2.2): $\sum_{n\le q} n^{-\sigma} \le \sum_{n\le q} n^{-\sigma_0} \le 1 + \int_1^q x^{-\sigma_0}\,\mathrm{d}x = 1 + (q^{1-\sigma_0}-1)/(1-\sigma_0) \le q^{1-\sigma_0}/(1-\sigma_0)$, and $|s\int_q^\infty S(x)x^{-s-1}\,\mathrm{d}x| \le |s|q\cdot q^{-\sigma}/\sigma \le |s|q^{1-\sigma_0}/\sigma_0$, since $q^{1-\sigma} \le q^{1-\sigma_0}$ and $1/\sigma \le 1/\sigma_0$. □

Lemma 2.2 (Borel–Carathéodory, [16, §5.5])**.** *Let f be analytic on $|z - z_0| \le R$ and $A = \max_{|z-z_0|=R}\operatorname{Re} f(z)$. Then for $0 < r < R$ and $|z - z_0| \le r$,*

$$|f(z)| \le \frac{2r}{R-r}(A + |f(z_0)|) + |f(z_0)|.$$

Proof. Apply the standard form $\max_{|z-z_0|\le r}|g| \le \frac{2r}{R-r}\max_{|z-z_0|=R}\operatorname{Re} g$ for $g = f - f(z_0)$, which has $g(z_0) = 0$ and $\max\operatorname{Re} g \le A + |f(z_0)|$. □

Lemma 2.3 (Perron kernel)**.** *For $c > 0$ and $x > 0$,* $\dfrac{1}{2\pi i}\displaystyle\int_{c-i\infty}^{c+i\infty}\frac{x^w\,\mathrm{d}w}{w(w+1)} = \left(1 - \frac{1}{x}\right)_+.$

Proof. The integrand is $O(x^{\operatorname{Re} w}|w|^{-2})$. For $x > 1$ move the line to $\operatorname{Re} w = -N$ and let $N \to \infty$; the residues at $w = 0$ and $w = -1$ are 1 and $-1/x$. For $0 < x < 1$ move the line to $\operatorname{Re} w = N$: no poles, and the integral tends to 0. For $x = 1$ the integral over $\operatorname{Re} w = N$ is $O(1/N)$, so the value is $0 = (1 - 1/x)_+$. □

Lemma 2.4 (Rosser–Schoenfeld)**.** *For $x > 1$, $\pi(x) < 1.25506\, x/\log x$ [14, (3.6)]. For $x \geq 285$,*

$$\prod_{p \leq x} \Big(1 - \frac{1}{p}\Big) > \frac{e^{-\gamma}}{\log x}\Big(1 - \frac{1}{2\log^2 x}\Big) \tag{2.3}$$

[14, Theorem 7, (3.25)]. (Dusart [17, Theorem 6.12] gives (2.3) with 0.2 in place of 1/2 for $x \geq 2973$; either version suffices below, since $x \geq e^{11}$ throughout.)

Lemma 2.5 (class number formula, [18, Chapter 6])**.** *For a fundamental discriminant $-d < 0$ let χ_{-d} be the Kronecker symbol, a real primitive character of conductor d, and $w \in \{2, 4, 6\}$ the number of roots of unity in $\mathbb{Q}(\sqrt{-d})$. Then $h(-d) = w\sqrt{d}\, L(1, \chi_{-d})/(2\pi) \geq \sqrt{d}\, L(1, \chi_{-d})/\pi$.*

2.2 Constants

Definition 2.6. For $1/2 \leq \theta < 1$ put $\delta = 1 - \theta$ and

$$a = \frac{22\delta}{25}, \quad \delta' = \frac{24\delta}{25}, \quad \sigma_0 = 1 - \delta', \quad \kappa = \frac{25}{\delta} + 22, \quad T_0 = 3 + 2\delta' + \frac{1}{\delta'},$$

$$A_1 = \kappa\delta', \quad A_2 = \kappa, \quad A_3 = \kappa \log \frac{1}{\sigma_0} + (\kappa + 1)\log \zeta(2),$$

$$K_0^+ = \frac{2\operatorname{arsinh}(1/a)}{1 - a} + 2, \qquad J^+ = 2\log(1 + T_0)\Big(\frac{\operatorname{arsinh}(1/a)}{1 - a} + 1 + \frac{1}{T_0}\Big),$$

where $\operatorname{arsinh} x = \log(x + \sqrt{x^2 + 1})$. For $M \geq 5$ and $q \geq 3$ put $\mathcal{L} = (M + \log\log q)/a$, $y = e^{\mathcal{L}}$,

$$E_0(q) = \frac{A_1 K_0^+ + (A_2 J^+ + A_3 K_0^+)/\log q}{2\pi e^M}, \qquad \Theta(q) = E_0(q) + \frac{1.25506}{\mathcal{L}} + \frac{2.5}{\sqrt{y}} + \frac{\mathcal{L}}{y^{2/3}} + \frac{1}{2\mathcal{L}^2 - 1} + \frac{2}{y}.$$

The geometric meaning: $R = 1 + \delta'$ and $r = 1 + a$ are the radii of the Borel–Carathéodory discs centred at $2 + it$, so that the outer disc lies in $\operatorname{Re} s \geq \sigma_0 = 1 - \delta' > \theta$ and the inner disc reaches $\operatorname{Re} s = 1 - a$; $\kappa = 2r/(R - r)$; $T_0 = 2 + 2R + \sigma_0/\delta'$. Since $a \leq 0.44$ and $M \geq 5$, one has $\mathcal{L} \geq (5 + \log\log 3)/0.44 > 11.5$ and $y > 10^5$. Every term of $\Theta(q)$ is decreasing in q ($\mathcal{L}/y^{2/3} = \mathcal{L}e^{-2\mathcal{L}/3}$ decreases for $\mathcal{L} \geq 3/2$), so $\Theta(q) \leq \Theta(3)$.

3 The Borel–Carathéodory bound

Proposition 3.1. *Under the hypotheses of Theorem 1.1,* (i) *holds.*

Proof. Fix $t \in \mathbb{R}$ and put $z_0 = 2 + it$, $R = 1 + \delta'$, $r = 1 + a$, so that $r < R$ and $2r/(R - r) = \kappa$. The closed disc $|s - z_0| \leq R$ lies in $\operatorname{Re} s \geq 2 - R = \sigma_0 = 1 - \frac{24}{25}\delta > \theta$, where $L(\cdot, \chi)$ is analytic (it is entire, χ being nonprincipal) and has no zeros by (H_θ). Hence $f(s) = \log L(s, \chi)$, defined on the disc by continuation from z_0 (where it equals the Dirichlet series), is analytic on the disc; the disc and the half-plane $\operatorname{Re} s > \theta$ are simply connected, so this f is the restriction of the branch fixed in Section 1.

Table 1: Constants of Theorem 1.1 for $M = 10$, rounded to the digits shown, with $C_- = e^{\gamma}e^{\Theta(3)}/(a\zeta(2))$ and $C_+ = e^{\gamma}e^{\Theta(3)}/a$; the constants 10.41, 17.12, 32.69 and 32.41 used in the text are upper bounds for the corresponding entries. Computed by `anc/constants.py`.

	$\theta = 7/8$	$\theta = \sigma_\dagger$ [3]	$\theta = 0.8739558$ [4]	$\theta = 1/2$ (GRH)
a	0.11	0.110038	0.110919	0.44
κ	222	221.93	220.34	72
T_0	11.5733	11.5706	11.5063	6.0433
A_1	26.64	26.641	26.662	34.56
A_3	139.366	139.334	138.581	83.415
K_0^+	8.5246	8.5241	8.5127	7.5691
J^+	22.0181	22.0150	21.9442	15.4215
$E_0(3)$	0.04160	0.04159	0.04120	0.01335
$\Theta(3)$	0.05534	0.05533	0.05505	0.06904
C_-	10.4034	10.3997	10.3142	2.6367
C_+	17.1129	17.1068	16.9662	4.3372
πC_-	32.6832	32.6715	32.4031	8.2835

On the disc, $\sigma_0 \le \sigma \le 2 + R$ and $|\operatorname{Im} s| \le |t| + R$, so $|s| \le |t| + 2 + 2R$. By (2.2) with this σ_0, using $1 - \sigma_0 = \delta'$,

$$|L(s,\chi)| \le q^{\delta'}\Big(\frac{1}{\delta'} + \frac{|t| + 2 + 2R}{\sigma_0}\Big) = \frac{q^{\delta'}}{\sigma_0}(|t| + T_0),$$

since $T_0 = 2 + 2R + \sigma_0/\delta'$. Therefore $A := \max_{|s-z_0|=R} \operatorname{Re} f(s) = \max \log|L(s,\chi)| \le \delta' \log q + \log(|t| + T_0) + \log(1/\sigma_0)$. At the centre, $|f(z_0)| \le \sum_{n\ge 2} \Lambda(n) n^{-2}/\log n = \log\zeta(2)$. Lemma 2.2 gives, for $|s - z_0| \le r$,

$$\begin{aligned} |\log L(s,\chi)| &\le \kappa\big(\delta' \log q + \log(|t| + T_0) + \log(1/\sigma_0) + \log\zeta(2)\big) + \log\zeta(2) \\ &= A_1 \log q + A_2 \log(|t| + T_0) + A_3. \end{aligned}$$

The segment $\{\sigma + it : 1 - a \le \sigma \le 3 + a\}$ is the diameter of the inner disc, which proves (i). □

4 The short Euler product

Lemma 4.1. *Let $0 < a < 1$, $T_0 > 0$, and for $v \in \mathbb{R}$ put $w = -a + iv$, so $|w| = \sqrt{a^2 + v^2}$ and $|w + 1| = \sqrt{(1-a)^2 + v^2}$. Then*

$$K_0 := \int_{-\infty}^{\infty} \frac{\mathrm{d}v}{|w||w+1|} \le K_0^+, \qquad J := \int_{-\infty}^{\infty} \frac{\log(|v| + T_0)\,\mathrm{d}v}{|w||w+1|} \le J^+.$$

Proof. For $|v| \le 1$ use $|w + 1| \ge 1 - a$ and $\int_{-1}^{1}(a^2 + v^2)^{-1/2}\,\mathrm{d}v = 2\operatorname{arsinh}(1/a)$; for $|v| > 1$ use $|w|, |w + 1| \ge |v|$, $\int_{|v|>1} v^{-2}\,\mathrm{d}v = 2$ and $\int_1^{\infty} \log(v + T_0)v^{-2}\,\mathrm{d}v = \log(1 + T_0) + \int_1^{\infty} \frac{\mathrm{d}v}{v(v+T_0)} = \log(1 + T_0)(1 + 1/T_0)$. For J on $|v| \le 1$ bound $\log(|v| + T_0) \le \log(1 + T_0)$. □

Lemma 4.2. *Let $y \ge 2$ and let χ be any Dirichlet character. Then*

$$\Big|\sum_{n\le y} \frac{\Lambda(n)\chi(n)}{n\log n}\Big(1 - \frac{n}{y}\Big) - \sum_{p\le y} -\log\Big(1 - \frac{\chi(p)}{p}\Big)\Big| \le \frac{\pi(y)}{y} + \frac{2.5}{\sqrt{y}} + \frac{\log y}{y^{2/3}}.$$

Proof. The first sum is $\sum_{p^k\le y}\chi(p)^k(1-p^k/y)/(kp^k)$ and the second is $\sum_{p\le y}\sum_{k\ge1}\chi(p)^k/(kp^k)$ (for $p\mid q$ both are 0). Their difference is

$$D=\sum_{p\le y}\sum_{k:\,p^k>y}\frac{\chi(p)^k}{kp^k}+\frac{1}{y}\sum_{p^k\le y}\frac{\chi(p)^k}{k}.$$

In the first term, if $p>\sqrt{y}$ then $\sum_{k\ge2}p^{-k}/k\le 1/(2p(p-1))$, and $\sum_{n>\sqrt{y}}1/(2n(n-1))=1/(2\lfloor\sqrt{y}\rfloor)\le 1/\sqrt{y}$; if $p\le\sqrt{y}$ and $k_0\ge3$ is the least k with $p^k>y$, then $\sum_{k\ge k_0}p^{-k}/k\le\frac{1}{k_0}\frac{p^{-k_0}}{1-1/p}<\frac{2}{3y}$, and there are at most $\sqrt{y}$ such p. So the first term is at most $5/(3\sqrt{y})$. In the second term, $\sum_{p^k\le y}1/k\le\pi(y)+\sum_{k=2}^{K}\pi(y^{1/k})/k$ with $K=\lfloor\log y/\log 2\rfloor$, and $\sum_{k=2}^{K}\pi(y^{1/k})/k\le\sqrt{y}/2+y^{1/3}\sum_{k=3}^{K}1/k\le\sqrt{y}/2+y^{1/3}\log y$. Adding, $|D|\le\pi(y)/y+5/(3\sqrt{y})+1/(2\sqrt{y})+\log y/y^{2/3}$, which is less than the stated bound. □

Proposition 4.3. *Under the hypotheses of Theorem 1.1,* (ii) *holds.*

Proof. Let $M\ge5$, $y=(e^M\log q)^{1/a}$, so $y^{-a}=1/(e^M\log q)$ and $y>10^5$. For $\operatorname{Re}w=1$ the series $\log L(1+w,\chi)=\sum_{n\ge2}\Lambda(n)\chi(n)n^{-1-w}/\log n$ converges absolutely, and $\sum_n\frac{\Lambda(n)}{n^2\log n}\int_{(1)}\frac{|y^w||\,\mathrm{d}w|}{|w||w+1|}<\infty$. By Lemma 2.3 with $x=y/n$ and Fubini,

$$\sum_{n\le y}\frac{\Lambda(n)\chi(n)}{n\log n}\Big(1-\frac{n}{y}\Big)=\frac{1}{2\pi i}\int_{(1)}\log L(1+w,\chi)\frac{y^w\,\mathrm{d}w}{w(w+1)}. \tag{4.1}$$

In the strip $-a\le\operatorname{Re}w\le1$ we have $\operatorname{Re}(1+w)\ge1-a>\theta$, so $\log L(1+w,\chi)$ is analytic there and the integrand has exactly one singularity, the simple pole at $w=0$ with residue $\log L(1,\chi)$ (the point $w=-1$ lies outside). On the horizontal segments $\operatorname{Im}w=\pm T$, $-a\le\operatorname{Re}w\le1$, the point $s=1+w$ has $1-a\le\sigma\le2$, so (i) gives $|\log L(1+w,\chi)|\le A_1\log q+A_2\log(T+T_0)+A_3$, while $|y^w|\le y$ and $|w(w+1)|\ge T^2$; these contributions tend to 0 as $T\to\infty$. Hence

$$\sum_{n\le y}\frac{\Lambda(n)\chi(n)}{n\log n}\Big(1-\frac{n}{y}\Big)=\log L(1,\chi)+E,\qquad E=\frac{1}{2\pi i}\int_{(-a)}\log L(1+w,\chi)\frac{y^w\,\mathrm{d}w}{w(w+1)}. \tag{4.2}$$

On $\operatorname{Re}w=-a$ write $w=-a+iv$; then $s=1+w=(1-a)+iv$ is in the range of (i) with $t=v$, and $|y^w|=y^{-a}$. Therefore

$$|E|\le\frac{y^{-a}}{2\pi}\int_{-\infty}^{\infty}\frac{A_1\log q+A_2\log(|v|+T_0)+A_3}{|w||w+1|}\,\mathrm{d}v=\frac{A_1K_0\log q+A_2J+A_3K_0}{2\pi e^M\log q}\le E_0(q)$$

by Lemma 4.1. Finally Lemma 4.2 replaces the left side of (4.2) by $\sum_{p\le y}-\log(1-\chi(p)/p)$ at a cost of $\pi(y)/y+2.5/\sqrt{y}+\log y/y^{2/3}$, and $\pi(y)/y<1.25506/\log y$ by Lemma 2.4. □

5 Proof of Theorem 1.1(iii) and of the corollaries

Proof of (iii). Take real parts in (ii): $\log|L(1,\chi)|=\sum_{p\le y}-\log|1-\chi(p)/p|+\operatorname{Re}\varepsilon$. Since $|\chi(p)|\le1$, $1-1/p\le|1-\chi(p)/p|\le1+1/p$, so

$$-\sum_{p\le y}\log\Big(1+\frac{1}{p}\Big)-|\varepsilon|\le\log|L(1,\chi)|\le-\sum_{p\le y}\log\Big(1-\frac{1}{p}\Big)+|\varepsilon|. \tag{5.1}$$

Write $\mathcal{L} = \log y \geq 11.5$. By (2.3) (valid since $y > 10^5$) and $-\log(1-x) \leq x/(1-x)$,

$$-\sum_{p\leq y} \log\Big(1-\frac{1}{p}\Big) < \gamma + \log\mathcal{L} - \log\Big(1-\frac{1}{2\mathcal{L}^2}\Big) \leq \gamma + \log\mathcal{L} + \frac{1}{2\mathcal{L}^2-1}. \tag{5.2}$$

For the lower bound use $1+1/p = (1-1/p^2)/(1-1/p)$, so that $\sum_{p\leq y}\log(1+1/p) = \log\prod_{p\leq y}(1-p^{-2}) - \sum_{p\leq y}\log(1-1/p)$. Now $\prod_{p\leq y}(1-p^{-2})^{-1} = \sum n^{-2}$ over the integers all of whose prime factors are at most y, which is at least $\sum_{n\leq y} n^{-2} \geq \zeta(2) - 1/\lfloor y\rfloor \geq \zeta(2) - 2/y$. Hence $\log\prod_{p\leq y}(1-p^{-2}) \leq -\log(\zeta(2)-2/y) \leq -\log\zeta(2) + 2/y$ (as $y \geq 4$), and with (5.2)

$$\sum_{p\leq y}\log\Big(1+\frac{1}{p}\Big) \leq -\log\zeta(2) + \gamma + \log\mathcal{L} + \frac{1}{2\mathcal{L}^2-1} + \frac{2}{y}.$$

Inserting this and (5.2) into (5.1), and using the bound for $|\varepsilon|$ of (ii), we obtain

$$\log\zeta(2) - \gamma - \log\mathcal{L} - \Theta(q) \leq \log|L(1,\chi)| \leq \gamma + \log\mathcal{L} + \Theta(q),$$

which is (iii) since $\mathcal{L} = (M + \log\log q)/a$. The monotonicity of Θ was noted after Definition 2.6. For real χ, $L(1,\chi)$ is real and positive (Dirichlet), so $|L(1,\chi)| = L(1,\chi)$. □

Proof of Corollary 1.2. For $\theta = 7/8$: $\delta = 1/8$, $a = 0.11$, $\delta' = 0.12$, $\sigma_0 = 0.88$, $\kappa = 222$, $T_0 = 11.5733\ldots < 11.58$, $A_1 = 26.64$, $A_2 = 222$, $A_3 = 222\log(1/0.88) + 223\log\zeta(2) = 139.366\ldots < 139.37$; the range $1-a \leq \sigma \leq 3+a$ is $0.89 \leq \sigma \leq 3.11$. This gives the bound on $\log L$. For $M = 10$, $\mathcal{L}(3) = (10+\log\log 3)/0.11 = 91.76\ldots$, $K_0^+ = 8.5246\ldots$, $J^+ = 22.0181\ldots$, $E_0(3) = 0.04160\ldots$, $1.25506/\mathcal{L}(3) = 0.01368\ldots$, and the remaining four terms of $\Theta(3)$ are below $6\cdot 10^{-5}$ in total, so $\Theta(3) = 0.05534\ldots$. Then $e^{\gamma}e^{\Theta(3)}/(0.11\,\zeta(2)) = 10.4034\ldots < 10.41$ and $e^{\gamma}e^{\Theta(3)}/0.11 = 17.1129\ldots < 17.12$. □

Proof of Corollary 1.3. For a fundamental discriminant $-d < 0$, χ_{-d} is a real primitive nonprincipal character modulo $d \geq 3$, so Corollary 1.2 and Lemma 2.5 give $h(-d) \geq \sqrt{d}/(\pi\cdot 10.4034\ldots(10+\log\log d))$ and $\pi\cdot 10.4034\ldots = 32.6832\ldots < 32.69$. The function $d \mapsto \sqrt{d}/(10+\log\log d)$ is increasing for $d \geq 3$ (its logarithmic derivative is $\frac{1}{2d} - \frac{1}{d\log d\,(10+\log\log d)} > 0$), so for each H the set of d with $\sqrt{d} > 32.69\,H(10+\log\log d)$ is an interval $[d_0(H),\infty)$, and $d_0(H)$ is computed by bisection in `anc/constants.py`: $d_0 = 1.6662\cdot 10^5$, $1.7548\cdot 10^7$, $1.8227\cdot 10^9$, $1.8780\cdot 10^{11}$ for $H = 1, 10, 100, 1000$. The last sentence of the corollary is proved in Section 6. □

6 Numerical section

6.1 Thresholds

For a given target H the parameter M may be chosen freely, and the error $\Theta(q)$ of Theorem 1.1(iii) decreases with q; both can be used when only large d matter. For each M the function $d \mapsto \sqrt{d}\,e^{-\Theta(d)}/(M+\log\log d)$ is increasing, so Theorem 1.1(iii) with $q = d$ gives $h(-d) > H$ for all $d \geq d_0(H;M)$, where $d_0(H;M)$ is the root of $\sqrt{d} = \pi C_-(d) H(M+\log\log d)$ with $C_-(d) = e^{\gamma}e^{\Theta(d)}/(a\zeta(2))$. Table 2 lists, for $\theta = 7/8$, the uniform thresholds (constant 32.69, $M = 10$) and the best thresholds over $M \in \{5, 5.25, \ldots, 15\}$; the optimum is $M \approx 6.5$–7, where $E_0(d) \approx 0.08$–0.1. The last two columns show what is known unconditionally.

The values 163, 13 843 and 2 383 747 are from Watkins's table [15] (all three were recomputed in `anc/checks.py` for $d \leq 3\cdot 10^6$); 1 062 350 083 is the largest discriminant with $h \leq 2000$ found by

Table 2: Thresholds $d_0(H)$ such that $h(-d) > H$ for all fundamental discriminants $-d < 0$ with $d \geq d_0(H)$, under the 7/8 half-plane (columns 2–3, this paper, rounded up; column 2 uses the constant 32.69), compared with the largest d with $h(-d) \leq H$ [15, 11] and with the unconditional thresholds of [11, Table 1].

H	uniform ($M = 10$)	best M	largest d, $h(-d) \leq H$	unconditional [11]
1	$1.67 \cdot 10^5$	$1.10 \cdot 10^5$ ($M = 7$)	163	–
10	$1.76 \cdot 10^7$	$1.13 \cdot 10^7$ ($M = 6.75$)	13 843	–
100	$1.83 \cdot 10^9$	$1.15 \cdot 10^9$ ($M = 6.75$)	2 383 747	$2.86 \cdot 10^{12}$
1000	$1.88 \cdot 10^{11}$	$1.17 \cdot 10^{11}$ ($M = 6.5$)	227 932 027	$3.46 \cdot 10^{14}$
2000	$7.58 \cdot 10^{11}$	$4.69 \cdot 10^{11}$ ($M = 6.5$)	1 062 350 083	$1.61 \cdot 10^{15}$
10^5	$1.97 \cdot 10^{15}$	$1.21 \cdot 10^{15}$ ($M = 6.5$)	unknown	–

Kimura [11, Theorem 2], attained at $h = 1978$, and 227 932 027 (attained at $h = 996$) is his value for $h \leq 1000$ [11, p. 3]; his preprint is not yet refereed, its analytic part comes with Lean proofs and its sieve computation does not. Kimura's unconditional thresholds come from explicit real-zero gaps $(1-\beta)\log q \geq c_N$ with $c_N \approx 0.0055$–0.0060 for $\log q \geq T_N$ [11, Table 1], made explicit from the argument of [12] (the uniform constant for all $q \geq 3$ is $5.8 \cdot 10^{-4}$); his Remark 11 states that under a half-plane $\operatorname{Re} s > 31/32$, which gives the gap 1/32, his Hecke–Landau-type bound with $c \approx 0.6$ yields $h(-d) > 2000$ for $d \geq e^{27.2} \approx 6.50 \cdot 10^{11}$ and $h(-d) > 10^5$ for $d \geq e^{35.4} \approx 2.37 \cdot 10^{15}$, and that the half-planes $\operatorname{Re} s > 7/8$ and $\operatorname{Re} s > 11/12$ give the same thresholds. At $d = e^{27.2}$ Corollary 1.3 gives only $h(-d) \geq 1853$, while the best choice of M gives $h(-d) \geq 2353$; at $d = e^{35.4}$ the two versions give $1.09 \cdot 10^5$ and $1.40 \cdot 10^5$. Thus in the range of current computations the Littlewood-type bound with the present constants is comparable to, not far better than, the explicit Hecke–Landau-type bound from the same input; its advantage, a denominator $\log\log d$ instead of $\log d$, becomes decisive only for larger d. For the finiteness statement "$h(-d) = H$ for at most finitely many d" it gives, under $(\mathrm{H}_{7/8})$, the explicit bound $d < d_0(H)$, where $\sqrt{d_0(H)} = 32.69\, H\,(10 + \log\log d_0(H))$, against the ineffective bound of Siegel [19] and the effective but logarithmic bounds of Goldfeld, Gross–Zagier and Oesterlé [20, 21, 22].

6.2 Why computation cannot test Corollary 1.3

The lower bound $\sqrt{d}/(32.69(10 + \log\log d))$ is below 1 for $d < 1.67 \cdot 10^5$, below 2 for $d < 6.7 \cdot 10^5$, and below 101 for $d < 1.83 \cdot 10^9$. The script `anc/checks.py` computes $h(-d)$ for all fundamental $d \leq 3 \cdot 10^6$ by counting reduced forms and verifies the corollary there (the minimum of $h(-d)$ divided by the bound is 26.1, at $d = 991\,027$). For $3 \cdot 10^6 < d < 1.83 \cdot 10^9$ one has $h(-d) \geq 101$ by Watkins's theorem [15], since all d with $h(-d) \leq 100$ are at most 2 383 747, while the bound is below 101. So the corollary is a theorem, unconditionally, for all $d < 1.83 \cdot 10^9$, and, relying on Kimura's determination of all d with $h(-d) \leq 2000$ [11] (not yet refereed), for all $d < 7.5 \cdot 10^{11}$. Numerical checks of the corollary are therefore only sanity checks of the constants, not tests of the input theorem. The same script checks Corollary 1.2 for all 164 562 nonprincipal characters modulo the primes $q \leq 1500$ (minimum of $|L(1,\chi)|$ divided by the lower bound: 29.7, attained at $q = 163$ by the real character; maximum of $|L(1,\chi)|$ divided by the upper bound: 0.019).

6.3 Comparison with other bounds

For $\theta = 1/2$ (GRH) Theorem 1.1 gives $L(1,\chi) \geq \zeta(2)e^{-\gamma} \cdot 0.44\, e^{-\Theta}/(M + \log\log q)$, against Littlewood's $(\frac{1}{2} + o(1))\zeta(2)e^{-\gamma}/\log\log q$; the loss is the factor 0.88 in the width, the additive M

and the factor $e^{-\Theta}$. The upper bound $17.12(10+\log\log q)$ of Corollary 1.2 improves on the trivial $|L(1,\chi)| \le \log q + 2$ (from (2.1) at $s=1$) only for $\log q > 264.7$; the content of the corollary is the lower bound. For primitive non-quadratic χ of conductor q the unconditional bound $|L(1,\chi)| \ge 1/(9.69030\log(q/\pi))$ of Mossinghoff, Starichkova and Trudgian [23] is stronger than Corollary 1.2 for $\log q < 14.78$, i.e. $q < 2.6\cdot 10^6$.

7 Proof of Theorem 1.4

The inputs are (2.1), Dirichlet's theorem $L(1,\chi)\neq 0$, and the representation

$$\zeta(s) = 1 + \frac{1}{s-1} - s\int_1^\infty \{x\}x^{-s-1}\,\mathrm{d}x \qquad (\sigma>0,\ s\neq 1), \tag{7.1}$$

where $\{x\} = x - \lfloor x\rfloor$; it follows by partial summation of $\sum n^{-s}$ against $\lfloor x\rfloor$ for $\sigma>1$ and analytic continuation (see [24, Chapter 1]).

Proof. Let $r = L(1,\chi)$ and $F(s) = \zeta(s)L(s,\chi) = \sum_{n\ge1} a_n n^{-s}$ for $\sigma>1$, $a_n = \sum_{d\mid n}\chi(d)$. As χ is real, $a_n = \prod_{p^k\| n}(1+\chi(p)+\cdots+\chi(p)^k) \ge 0$ and $a_1 = 1$.

Step 0: $r>0$. For real $s>1$, $F(s)\ge a_1 = 1$, while $\zeta(s)\to+\infty$ and $L(s,\chi)\to r$ as $s\to 1^+$; so $r\ge 0$, and $r\neq 0$ by Dirichlet's theorem.

Step 1. F is meromorphic on $\mathbb{C}$ with a single simple pole at $s=1$, of residue r, so $\Phi(s) = F(s) - r/(s-1)$ is entire. Expand $\Phi(s) = \sum_{k\ge0} c_k(2-s)^k$ at $s=2$. Since $(-1)^k\frac{\mathrm{d}^k}{\mathrm{d}s^k}\frac{1}{s-1}\big|_{s=2} = k!$, the pole term contributes $-r$ to each c_k, and the series contributes $b_k = \sum_n a_n(\log n)^k n^{-2}/k! \ge 0$ with $b_0 = F(2)\ge 1$. Thus $c_k = b_k - r$.

Step 2. On the circle $|s-2| = 3/2$ we have $\sigma\ge 1/2$, $|s|\le 7/2$ and $|s-1|\ge 1/2$. By (7.1), $|\zeta(s)| \le 1+2+\frac{7}{2}\cdot\frac{1}{\sigma}\le 10$. By (2.1), $|L(s,\chi)| \le \sum_{n\le q} n^{-1/2} + \frac{7}{2}q\cdot q^{-\sigma}/\sigma \le (2\sqrt{q}-1)+7\sqrt{q} \le 9\sqrt{q}$. Also, by (2.1) at $s=1$, $r\le \sum_{n\le q}1/n + 1 \le 2+\log q$, so $|r/(s-1)|\le 4+2\log q\le 10\sqrt{q}$ for $q\ge 3$. Hence $|\Phi(s)|\le 100\sqrt{q}$ on the circle and Cauchy's estimate gives $|c_k|\le 100\sqrt{q}\,(2/3)^k$.

Step 3. Let $\sigma = 1-\eta$, so $2-\sigma = 1+\eta < 3/2$ and $F(\sigma) = -r/\eta + \sum_{k\ge0}(b_k-r)(1+\eta)^k$. Put $K = \lceil 30\log q\rceil$. For the terms $k<K$, using $b_k\ge 0$, $b_0\ge 1$, $K\le 31\log q$ (as $\log q>1$) and $(1+\eta)^K \le e^{\eta K}\le e^{31/40} < 2.171$,

$$\sum_{k<K}(b_k-r)(1+\eta)^k \ge 1 - rK(1+\eta)^K \ge 1 - 2.171\cdot 31\, r\log q \ge 1 - 68\, r\log q.$$

For the tail, since $\eta\le 1/(40\log 3) < 1/8$ we have $(2/3)(1+\eta)\le 3/4$, so

$$\sum_{k\ge K}|b_k-r|(1+\eta)^k \le 100\sqrt{q}\sum_{k\ge K}(3/4)^k = 400\sqrt{q}\,(3/4)^K \le 400\, q^{1/2+30\log(3/4)} \le 400\cdot 3^{-8.13} < 0.06,$$

because $30\log(3/4) = -8.6304\ldots$ and the exponent $1/2+30\log(3/4) < -8.13$ is negative. Therefore

$$F(\sigma) \ge -40\, r\log q + 1 - 68\, r\log q - 0.06 > 0.9 - 108\, r\log q. \tag{7.2}$$

Step 4. Suppose $r < 1/(120\log q)$. Then $108\, r\log q < 0.9$ and $F(\sigma)>0$ by (7.2). By (7.1) with real $\sigma\in(0,1)$, $\zeta(\sigma) = \sigma/(\sigma-1) - \sigma\int_1^\infty\{x\}x^{-\sigma-1}\,\mathrm{d}x < 0$. Hence $L(\sigma,\chi) = F(\sigma)/\zeta(\sigma) < 0 < L(1,\chi)$, and the real continuous function $L(\cdot,\chi)$ vanishes somewhere in $(\sigma,1) = (1-\eta,1)$, contrary to the hypothesis. So $r\ge 1/(120\log q)$.

For the last assertions: if $\theta\le 1 - 1/(40\log 3)$ then $(1-\eta,1)\subset(\theta,1)$ for all $q\ge 3$, and the class-number bound follows from Lemma 2.5. ∎

8 Ancillary files

The directory `anc/` contains two scripts (Python 3, `mpmath`, `numpy`, `scipy`) and their outputs.

- `constants.py` evaluates Definition 2.6 for $\theta \in \{7/8, \sigma_\dagger, 0.8739558, 1/2\}$ and $M = 10$ (Table 1), the thresholds of Table 2 (uniform constant and best M), the comparison at Kimura's points $e^{27.2}$, $e^{35.4}$, and the numerical inequalities of Section 7 ($e^{31/40} < 2.171$, $400 \cdot 3^{1/2+30\log(3/4)} < 0.06$).
- `checks.py` verifies numerically: $K_0 \le K_0^+$ and $J \le J^+$ by quadrature; $\sum_{n \le q} n^{-\sigma_0} \le q^{\delta'}/\delta'$ at sample q; the prime-power count of Lemma 4.2 at $y = 10^5, 10^6, 10^7$; the two Rosser–Schoenfeld inequalities at sample $x \le 10^7$; the kernel identity; the class-number corollary for $d \le 3 \cdot 10^6$; and Corollary 1.2 for all characters modulo primes $q \le 1500$ via $L(1,\chi) = -q^{-1}\sum_a \chi(a)\psi(a/q)$.

The scripts check arithmetic and sanity; the proofs are in the text, and the input ($\mathrm{H}_{7/8}$) is not checked anywhere in this paper.

References

[1] OpenAI. The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\mathrm{Re}\, s > 7/8$, September 2026. OpenAI Math Release preprint, 30 September 2026, 199 pages. Repository https://github.com/openai/math, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Version: https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Quasi-Riemann-Hypothesis-September-30-2026/paper.pdf.

[2] OpenAI. The quasi-Riemann hypothesis: scope of the Lean formalization, 2026. Scope document `lean/docs/003.md`, repository https://github.com/openai/math, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/lean/docs/003.md.

[3] Jizhou Guo. A zero-free half-plane beyond seven eighths for Dirichlet L-functions and finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$, 2026. Preprint, 8 October 2026. https://github.com/aster2024/openai-math-followups/tree/main/zero-free-beyond-7-8.

[4] Quasi-Riemann Tracker. QRH Leaderboard, 2026. https://gtimg.github.io/QuasiRiemannTracker/; repository https://github.com/GTimG/QuasiRiemannTracker. Record for Dirichlet L-functions at the time of writing: 4369779/5000000 (J. Sussman, PR 86, 28 October 2026), not re-verified here.

[5] J. E. Littlewood. On the class-number of the corpus $P(\sqrt{-k})$. *Proc. London Math. Soc. (2)*, 27:358–372, 1928.

[6] Andrew Granville and K. Soundararajan. The distribution of values of $L(1, \chi_d)$. *Geom. Funct. Anal.*, 13:992–1028, 2003.

[7] John B. Friedlander. Certain hypotheses concerning L-functions. *Pacific J. Math.*, 69(1):37–44, 1977.

[8] Youness Lamzouri, Xiannan Li, and Kannan Soundararajan. Conditional bounds for the least quadratic non-residue and related problems. *Math. Comp.*, 84(295):2391–2412, 2015. Corrigendum: Math. Comp. 86 (2017), 2551–2554.

[9] Alessandro Languasco and Timothy S. Trudgian. Uniform effective estimates for $|L(1,\chi)|$. *J. Number Theory*, 236:245–260, 2022.

[10] OpenAI. The Quasi-Riemann Hypothesis, October 2026. OpenAI Math Release preprint, 5 October 2026 (zero-free half-plane $\operatorname{Re} s > 11/12$), 49 pages. Repository https://github.com/openai/math, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Version: https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Quasi-Riemann-Hypothesis-October-5-2026/paper2.pdf.

[11] Iwao Kimura. Imaginary quadratic fields of class number at most 2000, 2026. arXiv:2610.21511v2, 22 October 2026 (preprint, not yet refereed; Remark 11 is in v2 only). v1: 15 October 2026.

[12] OpenAI. Uniform exclusion of Landau–Siegel zeros, October 2026. OpenAI Math Release preprint, 1 October 2026, 9 pages. Repository https://github.com/openai/math, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, `preprints/Uniform-exclusion-of-Landau-Siegel-zeros-October-1-2026/paper.pdf`.

[13] Edmund Landau. Über die Klassenzahl imaginär-quadratischer Zahlkörper. *Nachr. Ges. Wiss. Göttingen, Math.-Phys. Kl.*, pages 285–295, 1918.

[14] J. Barkley Rosser and Lowell Schoenfeld. Approximate formulas for some functions of prime numbers. *Illinois J. Math.*, 6:64–94, 1962.

[15] Mark Watkins. Class numbers of imaginary quadratic fields. *Math. Comp.*, 73(246):907–938, 2004.

[16] E. C. Titchmarsh. *The Theory of Functions*. Oxford University Press, 2nd edition, 1939.

[17] Pierre Dusart. Estimates of some functions over primes without R.H., 2010. arXiv:1002.0442.

[18] Harold Davenport. *Multiplicative Number Theory*, volume 74 of *Graduate Texts in Mathematics*. Springer, 3rd edition, 2000.

[19] Carl Ludwig Siegel. Über die Classenzahl quadratischer Zahlkörper. *Acta Arith.*, 1:83–86, 1935.

[20] Dorian M. Goldfeld. The class number of quadratic fields and the conjectures of Birch and Swinnerton-Dyer. *Ann. Scuola Norm. Sup. Pisa Cl. Sci. (4)*, 3:624–663, 1976.

[21] Benedict H. Gross and Don B. Zagier. Heegner points and derivatives of L-series. *Invent. Math.*, 84:225–320, 1986.

[22] Joseph Oesterlé. Nombres de classes des corps quadratiques imaginaires. *Astérisque*, 121–122:309–323, 1985. Séminaire Bourbaki 1983/84, exp. 631.

[23] Michael J. Mossinghoff, Valeriia V. Starichkova, and Timothy S. Trudgian. Explicit lower bounds on $|L(1,\chi)|$, 2021. arXiv:2107.09230.

[24] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I. Classical Theory*, volume 97 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2007.

Use of AI

The proofs and the text of this paper were produced with AI systems (Claude Fable 5.1 and Claude Opus 5.5) under the author's direction. They were checked by independent reviews with AI models and by the computations in the ancillary files. No human expert has reviewed them. The author takes responsibility for the content.